242407: metadacite, Bungeye Well

(Mount Gibson Formation, Murchison Supergroup, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Ninghan (SH 50-7), Mount Gibson (2338)
MGA Zone 50, 514200E 6730621N

Sampled on 20 May 2019

This sample was collected from the 187.2 – 187.9 m depth interval of diamond drillcore MGD018, drilled in 2009 by Accent Resources NL (Allen, 2010). The drillhole is located west of the Great Northern Highway on Ninghan Station, about 9.0 km west-southwest of Yandhanoo Hill, 6.3 km northwest of the top of Mount Gibson, and 1.9 km east-southeast of Bungeye Well.

Geological context

The unit sampled is a metadacite assigned to the 2934–2929 Ma Mount Gibson Formation of the Murchison Supergroup (Van Kranendonk et al., 2013). The Mount Gibson Formation consists mainly of metamorphosed felsic volcanic and volcaniclastic rocks of rhyolitic to dacitic composition (Ivanic, 2021). The oldest recognized felsic metavolcanic rocks in the western Youanmi Terrane, represented by the Madoonga, Gossan Hill, and Mount Gibson Formations, are coeval with 3010–2930 Ma metagranitic to metatonalitic rocks of the Thundelarra Supersuite (Van Kranendonk et al., 2013). A metamonzogranite of the Thundelarra Supersuite, sampled about 23 km to the south, yielded an igneous crystallization age of 2928 ± 5 Ma (GSWA 242405, Wingate et al., 2021). Another metamonzogranite of the Thundelarra Supersuite, collected about 86 km to the northwest, yielded an igneous crystallization age of 2946 ± 4 Ma (GSWA 219311, Wingate et al., 2020).

Petrographic description

The sample is a fine-grained, weakly laminated metadacite (Fig. 1), consisting of about 98–99% groundmass, 1% quartz phenocrysts, <1% plagioclase phenocrysts, and minor apatite and zircon. The groundmass consists of subequal proportions of quartz, feldspar and biotite. Quartz and feldspar form a semi-polygonal to polygonal aggregate with an average grain size of 0.1 mm. Within this aggregate, plagioclase exhibits sericite alteration and rare twinning, and some K-feldspar displays microcline twinning. Biotite forms mainly anhedral basal sections and rare semi-aligned subhedral laths up to 0.25 mm in size. Some biotite has been replaced by chlorite. Apatite forms euhedral laths up to 0.1 mm long. A weak, slightly wavy lamination is defined by discontinuous concentrations of biotite. Quartz phenocrysts are anhedral, equant to elongate, up to 2 mm in size, and have smooth curved boundaries. All have been recrystallized to a microcrystalline (mostly 0.1 – 0.5 mm) aggregate of polygonal and interlocking unstrained grains. Plagioclase phenocrysts are anhedral, slightly tabular, pitted, and up to 0.6 mm in size. The weak alignment of biotite suggests that the lamination is remnant flow banding rather than a tectonic foliation.

Zircon morphology

Zircons isolated from this sample are pale brown to dark brown, and subhedral to euhedral. The crystals are up to 350 µm long, and equant to elongate, with aspect ratios up to 5:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 4–5 September 2020, using SHRIMP-B. Nine analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.21% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.55%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Sixteen analyses were obtained from 16 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to slightly discordant (Fig. 3), and form a single group, based on their $^{207}Pb^{*}/^{206}Pb^{*}$ ratios.

Group I comprises 16 analyses (Table 1), which yield a weighted mean $^{207}Pb^{*}/^{206}Pb^{*}$ date of 2932 ± 5 Ma (MSWD = 0.72).

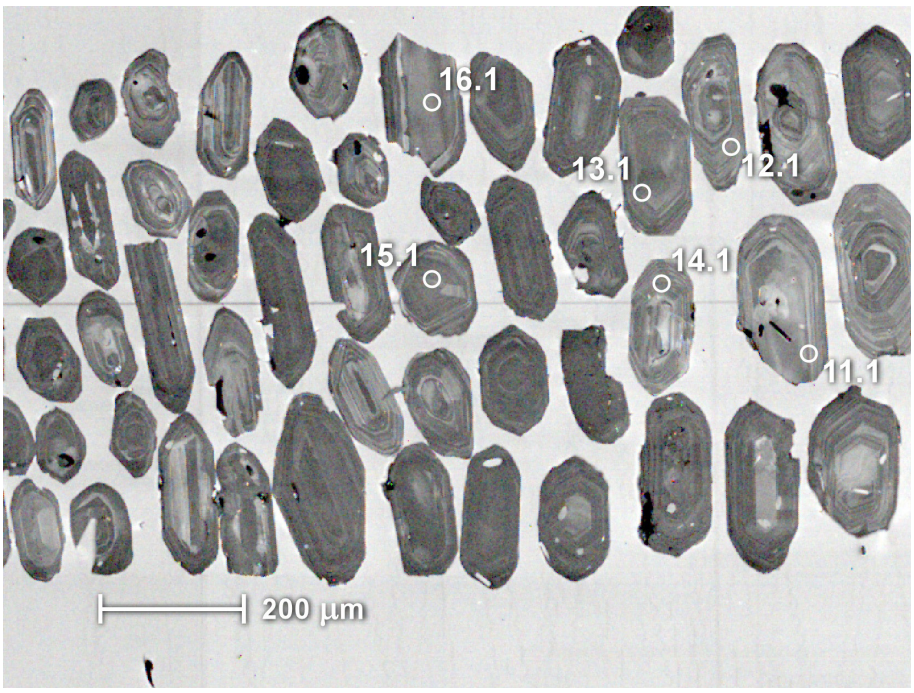

Figure 1. Drillcore image of sample 242407: metadacite, Bungeye Well

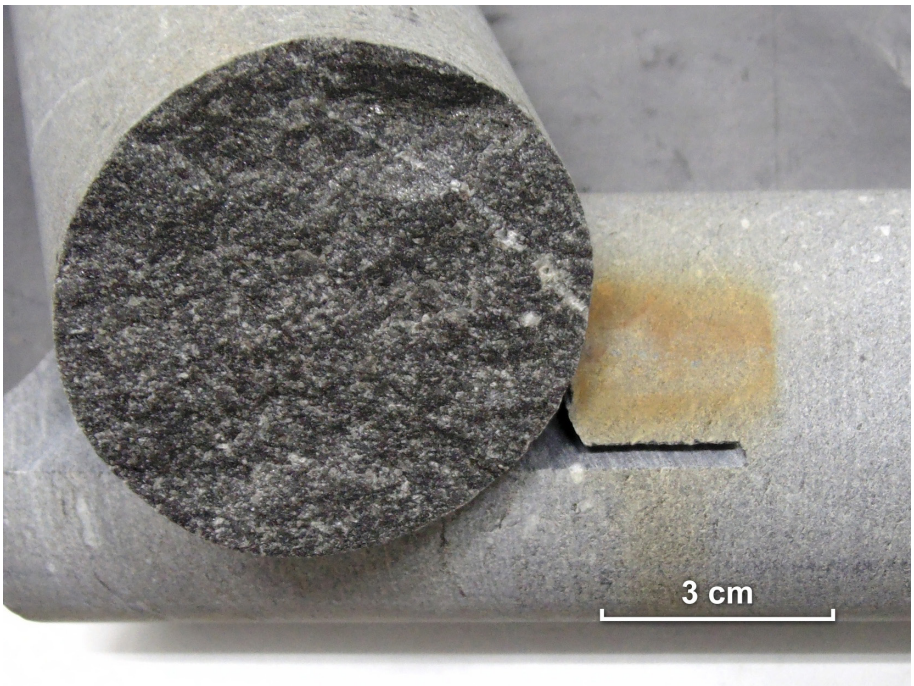

Figure 2. Cathodoluminescence image of representative zircons from sample 242407: metadacite, Bungeye Well. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 242407: metadacite, Bungeye Well

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	3	3.1	120	72	0.60	0.061	1.732	0.020	0.21223	0.00139	1.733	0.020	0.21168	0.00142	2936	28	2918	11	-0.6
I	11	11.1	118	80	0.67	0.016	1.737	0.021	0.21211	0.00140	1.738	0.021	0.21197	0.00141	2931	28	2921	11	-0.3
I	2	2.1	62	35	0.57	0.000	1.741	0.026	0.21213	0.00180	1.741	0.026	0.21213	0.00180	2926	36	2922	14	-0.1
I	7	7.1	120	89	0.74	0.047	1.761	0.021	0.21298	0.00139	1.761	0.021	0.21256	0.00141	2899	27	2925	11	0.9
I	8	8.1	202	79	0.39	-0.009	1.793	0.018	0.21254	0.00115	1.793	0.018	0.21262	0.00115	2857	23	2926	9	2.3
I	12	12.1	209	120	0.58	0.019	1.727	0.017	0.21306	0.00118	1.728	0.017	0.21289	0.00119	2944	24	2928	9	-0.6
I	16	16.1	132	42	0.32	0.029	1.776	0.020	0.21336	0.00135	1.777	0.020	0.21310	0.00137	2878	27	2929	10	1.7
I	13	13.1	175	114	0.65	0.000	1.759	0.018	0.21340	0.00179	1.759	0.018	0.21340	0.00179	2902	25	2932	14	1.0
I	1	1.1	131	76	0.58	0.000	1.715	0.020	0.21341	0.00138	1.715	0.020	0.21341	0.00138	2962	27	2932	10	-1.0
I	14	14.1	177	100	0.57	0.060	1.773	0.018	0.21408	0.00120	1.774	0.018	0.21354	0.00122	2882	24	2933	9	1.7
I	5	5.1	134	70	0.52	0.014	1.770	0.020	0.21391	0.00135	1.770	0.020	0.21379	0.00136	2887	26	2934	10	1.6
I	9	9.1	112	71	0.64	0.033	1.791	0.023	0.21435	0.00143	1.791	0.023	0.21406	0.00145	2859	29	2937	11	2.6
I	15	15.1	182	104	0.57	0.000	1.793	0.018	0.21480	0.00124	1.793	0.018	0.21480	0.00124	2857	24	2942	9	2.9
I	10	10.1	185	108	0.58	0.021	1.738	0.018	0.21534	0.00122	1.739	0.018	0.21515	0.00123	2929	24	2945	9	0.5
I	4	4.1	185	72	0.39	-0.011	1.764	0.018	0.21522	0.00123	1.763	0.018	0.21532	0.00123	2896	24	2946	9	1.7
I	6	6.1	169	138	0.81	0.011	1.754	0.018	0.21612	0.00304	1.754	0.018	0.21602	0.00304	2908	25	2951	23	1.5

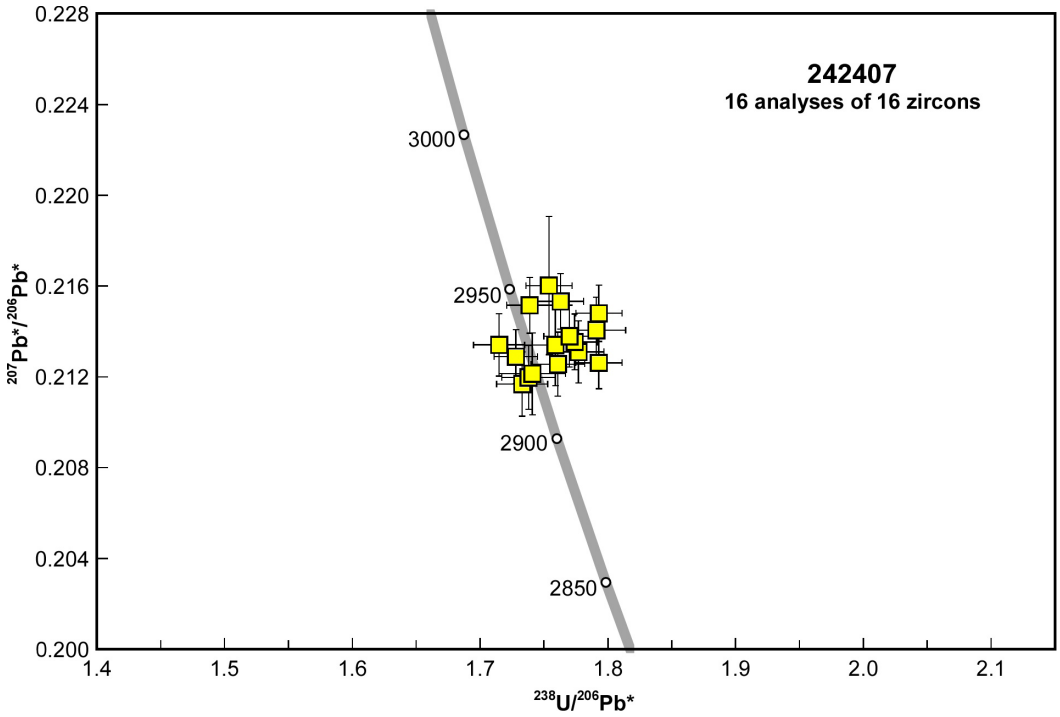

Figure 3. U–Pb analytical data for sample 242407: metadacite, Bungeye Well. Yellow squares indicate Group I (magmatic zircons)

The date of 2932 ± 5 Ma for the 16 analyses in Group I is interpreted as the igneous crystallization age of the dacite.

References

Allen, D 2010, Annual Report for Exploration Licence E59/875, Magnetite Range for the Period Ending 21 March 2010: Annual Exploration Report, 1200p. (unpublished).

Ivanic, TJ (partial report), Mount Gibson Formation (A-MUb-xfr-fd): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 30 July 2021, <www.dmp.wa.gov.au/ens>.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Van Kranendonk, MJ, Ivanic, TJ, Wingate, MTD and Kirkland, CL 2013, Long-lived, autochthonous development of the Archean Murchison Domain, and implications for Yilgarn Craton tectonics: Precambrian Research, v. 229, p. 49–92.

Wingate, MTD, Fielding, IOH, Lu, Y and Ivanic, T 2021, 242405: metamonzogranite, Mount Gibson mine; Geochronology Record 1805: Geological Survey of Western Australia, 4p.

Wingate, MTD, Lu, Y and Zibra, I 2020, 219311: metamonzogranite, North Paradise Well; Geochronology Record 1680: Geological Survey of Western Australia, 4p.

Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Ivanic, T 2021, 242407: metadacite, Bungeye Well; Geochronology Record 1806: Geological Survey of Western Australia, 4p.

Data obtained: 5 September 2020

Data released: 21 October 2021